

CSCI 380.61E Web Programming and Interface Design

Part 3

Final Project & Documentation

Objectives

- Deliver a polished, functional web application.
- Prepare complete documentation and materials.

1. Project Requirements

Your task is to build a complete, data-driven web application. The architecture must strictly follow the requirements below:

- Frontend: Built with React.js. You must use functional components and hooks (e.g., `useState`, `useEffect`) for state management.
- Backend: An API server built using HapiJS or ExpressJS.
- Database: Persistent storage using MongoDB (local or Atlas) or MySQL.
- Communication: Proper RESTful communication via HTTP requests (GET, POST, PUT, DELETE) facilitating full CRUD functionality.
- Testing: Implementation of automated testing for both logic and endpoints.

2. Report & Documentation

The final report must be professionally structured and include:

I. Project Background & Functions

- Problem Statement: What real-world problem does this website solve?
- System Design (architecture diagram, data flow)
- Functional Requirements: A detailed list of what the user can do (e.g., "Users can register," "Users can post reviews").

II. Technical Specification

- API Endpoint Descriptions: A clear table documenting your routes.

I Method	I Endpoint	I Description	I Request Body Example
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I POST	I /api/users/register	I Creates a new user	I { "user": "...", "pass": "..." } I
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I GET	I /api/products/:id	I Fetches details for a specific item	I N/A I
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- Database Schema: Description of your MongoDB collections, fields, and data types.

III. Testing Evidence (Screenshots)

- Website Walkthrough: Key pages of your live React application.
- API Verification: Proof of API functionality using Postman or cURL output.
- Automated Tests: Successful test runs using Jest (recommended for React) or Mocha (recommended for backend).

3. Submission

Note: Failure to host the website online will result in a pending grade, as the application must be accessible for final evaluation.

- Source Code: A .zip file containing the clean project (exclude node_modules).
- GitHub Repository: A link to your public/private repo (ensure the instructor has access).
- Live URL: Link to the hosted website (e.g. AWS EC2).
- Final Report: A PDF document covering all sections listed above.